

DevPilot Test Results

Consolidated report through 2 October 2026

This report brings together the recorded project checks, retrieval evaluations, local-model answer tests, repair and container runs, local adapter experiment, and the latest public-repository test. It separates retrieval scores from answer quality and states where a test was only a small diagnostic.

Executive findings

Area	Recorded result	What it means
Project tests	96 passed, 1 skipped	The skipped test depends on Docker. Frontend formatting and production build passed at the recorded checkpoint.
Repository retrieval	81.2% recall@8 on 48 development questions; 93.8% on 16 older holdout questions	The older split is contaminated by prior development and symbol-seeded labels; it is not a clean generalization score.
Multi-stage retrieval	Quick 37.5%; Deep 54.2% on four questions	Deep helped this tiny workflow set, but mean response time rose from 269 ms to 611 ms.
Generated answers	Correct citation syntax did not guarantee support or completeness	Manual review found wrong-source citations, incomplete answers, and unsupported statements.
Local pattern training	80-step adapter completed; it did not improve four held-out repository answers	The adapter remains experimental and is not enabled in DevPilot.
Latest public repositories	6 of 9 expected symbols retrieved; all three answer reviews found a material limitation	Useful diagnostic evidence, not a broad benchmark.

How to read these results

Recall@8 measures whether labelled symbols appear among the eight retrieved results. It does not score whether a generated explanation is correct. Citation checks mostly verify citation numbers and placement; they do not prove entailment. Repo-specific test runs only cover the commands and selected tests named in the record.

Models were run locally through Ollama. No cloud model or cloud training service was used for these records.

Retrieval and index evaluations

Eight reference repositories

The original real-repository check used three symbol questions per repository (24 total). Hybrid recall@8 averaged 67%; the small sample missed descriptive questions in HTTPX and pytest.

Repository	Indexed files	Symbols	Hybrid recall@8
Flask	183	1,805	67%
Requests	64	871	100%
HTTPX	90	1,331	33%
pytest	647	7,975	33%
Rich	504	2,600	100%
Spring Petclinic	59	299	67%
Commons Lang	659	12,985	67%
Mockito	1,017	9,985	67%

Expanded retrieval set

The latest 72-case file contains 64 answerable symbol questions plus eight external-state questions. On answerable cases, hybrid recall@8 was 81.2% (MRR 0.580) for 48 development questions and 93.8% (MRR 0.618) for 16 older, contaminated holdout questions. Many labels were created with the target symbol already known, so these scores primarily test symbol lookup.

The separate 12-question Click/JUnit 4 checkpoint reached 83.3% hybrid recall@8 and 0.38 MRR after tuning, up from 75% and 0.34 on its first scored checkpoint. It is small and must not be treated as an untouched statistical holdout.

LangGraph and workflow retrieval

Set	Quick recall@8	Deep recall@8	Mean time
4 multi-stage workflow questions	37.5%	54.2%	269 ms / 611 ms
64 mostly symbol-seeded questions	84.4%	84.4%	267 ms / 313 ms
12 Click/JUnit 4 checkpoint questions	83.3%	83.3%	175 ms / 211 ms

These runs did not call an LLM. A four-question workflow probe exposed weak cross-file retrieval, especially for the pytest workflow. Reindexing the same eight reference commits added files and configuration links but left the 48-case retrieval score unchanged.

Answer and model tests

Local model comparison and citation review

On four first-question prompts, Llama 3.2 produced numbered citations in 0/4 answers at a 16.5-second mean. Qwen 2.5 Coder 7B did so in 2/4 at a 33-second mean. After the citation prompt was tightened, a two-question repeat produced cited output in 2/2 Llama answers and 1/2 Qwen answers. These counts measure citation form, not accuracy, and are too small to rank the models.

In two Flask answer-review smokes, one answer confused Flask and View dispatch; the other had a correct conclusion but used a citation that did not support every claim. A Qwen repeat produced one correct answer with citation-support gaps and one fallback to source locations. Later schema checks improved formatting but still did not establish semantic support.

Pydantic Quick and Deep test

Question	Quick recall	Deep recall	Answer review
Easy: model_dump	1/1	1/1	Both explanations correctly name the serializer call and exclude_none forwarding.
Medium: model_json_schema	2/3	2/3	Quick makes an unsupported mock-schema rebuild claim; Deep omits the exact rebuilding behavior.
Hard: unresolved forward reference	3/6	6/6	Deep retrieves all expected symbols and gives a lifecycle explanation; deterministic relevance checks do not prove entailment.

All six responses were generated without provider warnings. This three-question diagnostic is not a general accuracy estimate. No Pydantic project tests or container runs were executed.

Abstention test

Eight reviewed questions about live, private, or future values were all answered with an abstention after a deterministic external-state guard was added (8/8). The first two had elicited invented model answers before the guard. This is a narrow guard contract, not a general hallucination rate.

Patch, container, and training tests

Code execution and repair

Run	Result	Scope
Python parcel quote demo	Baseline failed (exit 1); patched run passed all 4 tests (exit 0).	One demo regression only.
Commons Lang synthetic fixture	1/1 repair task verified; test failed before and passed after the hand-authored patch.	Synthetic fixture, not a real upstream issue.
Java runners in offline containers	Petclinic OwnerControllerTests, Commons Lang StringUtilsTest, and Mockito MockUtilTest passed.	One selected test command per repository.
Java generated test fixture	Distinctive failing-before/passing-after transition confirmed.	Fixture-only; not a model-authored repository repair.
Qwen BitField repair attempt	Generated expectation for clear(0) was wrong; test failed before and after patch.	Failed repair; no regression demonstrated.
Qwen Java edit attempts	Rejected for mismatched edits, timeout, and a test path absent from the pinned index.	No valid end-to-end model-authored repair.
LangGraph guided repair	Controlled fixture completed review/checkpoint/container flow. A Qwen draft contradicting existing zero-weight validation was declined before Docker.	Orchestration validated; real model-authored repair remains unproven.

Local transferable-pattern training

A local LoRA adapter was trained on repository-tree metadata from 15 pinned repositories (76 training examples, 15 validation examples, 4 held-out transfer cases). Source-code bodies were excluded. Across 80 iterations, validation loss fell from 2.764 to 0.733, with about 6.03 GB reported peak memory.

The lower validation loss did not transfer: the final adapter was repetitive, omitted concrete paths, and emitted malformed end markers on the four held-out trees. Its mean generation time was 19.44 seconds versus 18.27 seconds for the base checkpoint. The 20-step checkpoint was more specific but still made unsupported inferences. The adapter is not integrated or enabled.

Latest public repository test

The same evidence-focused question was run through Quick and Deep on each pinned repository using local Qwen 2.5 Coder 7B. The test uses the project's repository snapshots; it does not run those projects' own test suites.

Test tier and repository	Indexed scope	Recall@8	Manual answer finding
Medium: FastAPI	Repository root; 2,879 files, 8,871 symbols	1/2 (50%)	Found FastAPI.openapi but missed get_openapi. Delegation was described, but schema assembly was not established by the retrieved evidence.
Complex: Django	django/ only; 1,114 files, 12,728 symbols	3/3 (100%)	Found all expected methods. Citation [4] points to RawQuerySet._fetch_all, while the answer names QuerySet._fetch_all, so the claim is not properly supported by that citation.
Very complex: PyTorch	torch/ only; 2,597 files, 60,945 symbols	2/4 (50%)	Found _call_impl and pre-hook registration, but missed wrapper and forward-hook registration. The answer stopped before explaining forward() and post-hooks.

Quick and Deep returned the same evidence and answer for each case; all Deep runs used one retrieval pass. All six outputs passed citation-format checks, but the checks do not establish claim support. Semantic embeddings were not active for this run.

Current conclusion

- Project tests, selected Java runner tests, and controlled patch-flow fixtures pass within their stated scope.
- Symbol retrieval performs better on target-seeded cases than on multi-stage questions; Deep retrieval adds cost and its gain is inconsistent.
- Generated answers need human source review. Citation syntax and relevance checks do not prove a claim is true or fully supported.
- The local pattern adapter did not improve unseen-repository answers and should remain unused.

Primary result records

Project health: docs/progress.md; original repository set: docs/evaluation.md and docs/evaluation-results.json; expanded retrieval: evaluation/expanded-retrieval-v3.json; LangGraph: docs/langgraph.md and evaluation/langgraph-*-response-checkpoint.json; Pydantic: docs/pydantic-question-test.md and evaluation/pydantic-devpilot-question-run-v14.json; answer and repair checks: docs/model-comparison.md, docs/model-validation*.json, docs/container-validation.json, docs/java-*-validation.json, evaluation/repairs/fixture-commons/result.json; adapter: docs/local-training.md and evaluation/transfer-training/RESULTS.md; newest repo test: evaluation/public-repository-tier-test.md and evaluation/public-repository-tier-test.json.